PHYS 143 – Mechanics and Thermal Physics

Week 1: Vectors – Supplementary Notes

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These notes support the Week 1 lecture. They are not covered in Young and Freedman, Chapter 1, and they are not required for the Week 1 lecture problems. Read them after the lecture.

1 Linear independence of the unit vectors

In the lecture we stated that the mutually perpendicular unit vectors $\hat{\boldsymbol{\imath}}$, $\hat{\boldsymbol{\jmath}}$ and $\hat{\boldsymbol{k}}$ provide three independent directions, so a vector is uniquely specified by its x-, y- and z-components. This section states that idea precisely.

Definition: linear independence

A set of vectors $\vec{\boldsymbol{u}}_1, \vec{\boldsymbol{u}}_2, \ldots, \vec{\boldsymbol{u}}_n$ is **linearly independent** if the only way to combine them to give the zero vector is with every coefficient equal to zero:

$$c_1\vec{\boldsymbol{u}}_1 + c_2\vec{\boldsymbol{u}}_2 + \cdots + c_n\vec{\boldsymbol{u}}_n = \vec{\mathbf{0}} \quad \Longrightarrow \quad c_1 = c_2 = \cdots = c_n = 0.$$

The unit vectors are linearly independent. Suppose $a\hat{\boldsymbol{\imath}} + b\hat{\boldsymbol{\jmath}} + c\hat{\boldsymbol{k}} = \vec{\mathbf{0}}$. Take the dot product of both sides with $\hat{\boldsymbol{\imath}}$. Because $\hat{\boldsymbol{\imath}} \cdot \hat{\boldsymbol{\imath}} = 1$ and $\hat{\boldsymbol{\imath}} \cdot \hat{\boldsymbol{\jmath}} = \hat{\boldsymbol{\imath}} \cdot \hat{\boldsymbol{k}} = 0$, this gives $a = 0$. Dotting with $\hat{\boldsymbol{\jmath}}$ and $\hat{\boldsymbol{k}}$ in the same way gives $b = 0$ and $c = 0$.

Geometrically, no combination of $\hat{\boldsymbol{\imath}}$ and $\hat{\boldsymbol{\jmath}}$ can produce $\hat{\boldsymbol{k}}$, because any combination $a\hat{\boldsymbol{\imath}} + b\hat{\boldsymbol{\jmath}}$ lies in the xy-plane, while $\hat{\boldsymbol{k}}$ is perpendicular to that plane.

Consequence: components are unique

If $\vec{\boldsymbol{A}} = A_x\hat{\boldsymbol{\imath}} + A_y\hat{\boldsymbol{\jmath}} + A_z\hat{\boldsymbol{k}}$ and also $\vec{\boldsymbol{A}} = A'_x\hat{\boldsymbol{\imath}} + A'_y\hat{\boldsymbol{\jmath}} + A'_z\hat{\boldsymbol{k}}$, then subtracting gives

$$(A_x - A'_x)\hat{\boldsymbol{\imath}} + (A_y - A'_y)\hat{\boldsymbol{\jmath}} + (A_z - A'_z)\hat{\boldsymbol{k}} = \vec{\mathbf{0}},$$

so $A_x = A'_x$, $A_y = A'_y$ and $A_z = A'_z$. Hence two vectors are equal if and only if their corresponding components are equal:

$$\vec{\boldsymbol{A}} = \vec{\boldsymbol{B}} \iff A_x = B_x, \quad A_y = B_y, \quad A_z = B_z.$$

This is why a single vector equation can be split into separate x, y and z equations, a technique used throughout mechanics (for example, when resolving Newton's second law into components).

2 Extension: Direction Angles in 3D

In two dimensions a single angle θ specifies a direction. In three dimensions we can instead give the three angles α, β and γ that the vector $\vec{\boldsymbol{A}}$ makes with the positive x-, y- and z-axes. Their cosines are called the **direction cosines**.

Using the dot product with each unit vector, $\vec{\boldsymbol{A}} \cdot \hat{\boldsymbol{\imath}} = A\cos\alpha$ and also $\vec{\boldsymbol{A}} \cdot \hat{\boldsymbol{\imath}} = A_x$. Therefore:

Rule: direction cosines

$$\cos\alpha = \frac{\vec{\boldsymbol{A}}\cdot\hat{\boldsymbol{\imath}}}{A} = \frac{A_x}{A}, \qquad \cos\beta = \frac{\vec{\boldsymbol{A}}\cdot\hat{\boldsymbol{\jmath}}}{A} = \frac{A_y}{A}, \qquad \cos\gamma = \frac{\vec{\boldsymbol{A}}\cdot\hat{\boldsymbol{k}}}{A} = \frac{A_z}{A}$$

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

The identity follows directly: $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = (A_x^2 + A_y^2 + A_z^2)/A^2 = 1$. The direction cosines are also the components of the unit vector along $\vec{\boldsymbol{A}}$:

$$\hat{\boldsymbol{A}} = \frac{\vec{\boldsymbol{A}}}{A} = \cos\alpha\,\hat{\boldsymbol{\imath}} + \cos\beta\,\hat{\boldsymbol{\jmath}} + \cos\gamma\,\hat{\boldsymbol{k}}.$$

Worked example

Find the direction angles of $\vec{\boldsymbol{A}} = \hat{\boldsymbol{\imath}} + 9\hat{\boldsymbol{\jmath}} + 2\hat{\boldsymbol{k}}$ (the vector used in Contact 3), and check the identity.
Magnitude: $A = \sqrt{1^2 + 9^2 + 2^2} = \sqrt{86} = 9.27$.
Direction angles:

$$\alpha = \cos^{-1}\frac{1}{9.27} = 83.8^\circ, \qquad \beta = \cos^{-1}\frac{9}{9.27} = 13.9^\circ, \qquad \gamma = \cos^{-1}\frac{2}{9.27} = 77.5^\circ.$$

Check: $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \dfrac{1 + 81 + 4}{86} = 1.$
Evaluate: β is small, so $\vec{\boldsymbol{A}}$ points mostly along $+y$, which agrees with its large y-component.

Use. Direction angles provide a compact way of describing the orientation of a vector in three dimensions.

Exercise. Find the direction angles of $\vec{\boldsymbol{B}} = -5\hat{\boldsymbol{\imath}} + \hat{\boldsymbol{\jmath}} - \hat{\boldsymbol{k}}$. (Answer: $\alpha = 164.2^\circ$, $\beta = 78.9^\circ$, $\gamma = 101.1^\circ$.)